Lab Session 1

Getting Oriented; Practicing With Micropipettes

Goal: Starting next week, you will be working on a laboratory project that will build throughout the entire semester. Before embarking on that journey, it is important to familiarize yourself with your lab space and to master the use of the work-horses of the molecular biology/biochemistry lab: the micropipettes. If your instructor has not a given safety orientation yet, they will do so today.

I. EQUIPMENT CHECKLIST

It is important to familiarize yourself with the work environment and laboratory equipment before beginning experiments. If the laboratory space in which you are working is shared by other laboratory sections at different times, much of the equipment can be shared. There are certain items, however, such as buffers and sterile disposable items that should not be shared between lab groups. Take a moment to go through your bench, shelves, and drawers to identify equipment and reagents. Use the checklist below and notify your instructor if anything is missing. Items that are indicated as “per group” should not be shared between different sets of students on different lab days. Label these items with your initials and lab day.

- One power supply box
- One horizontal DNA mini gel apparatus for agarose gels
- Four micropipettes: P10, P20, P200, and P1000
- One box 1000 µL sterile tips *per group*
- One box 200 µL sterile tips *per group*
- One box 10 µL sterile tips *per group*
- One ice bucket (or cooler or styrofoam box for ice)
- One box of Kimwipes (Kimberly-Clark, Roswell, GA)
- One 15 mL and one 50 mL styrofoam test tube rack
- One pack of sterile snap cap tubes (17 × 100 mm) for overnight bacterial cultures
- One test tube rack for snap cap tubes
- One autoclaved container of 1.7 mL microcentrifuge tubes *per group*
- Two microcentrifuge tube racks
- One pack of disposable 10 mL pipettes
- One plastic (or electric) pipette pump
- One 50 mL graduated cylinder
- One 500 mL graduated cylinder
- One 2 L polypropylene beaker
- Two 1 L polypropylene bottles, one for deionized water and one for 1 Tris/Borate/EDTA (TBE) × buffer, *per group*
- One 250 or 500 mL Pyrex orange capped bottle or Erlenmeyer flask for melting agarose *per group*
- One thermal glove for handling microwaved agarose
- Labeling tape
- Permanent ink marker (Sharpie)
- One plastic squeeze bottle for 70% ethanol
- One plastic squeeze bottle for deionized water
- One ring stand with clamp
- One pair of blunt-ended forceps
- Two safety eye glasses or goggles
- One cardboard freezer box *per group*

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- Protein polyacrylamide mini gel electrophoresis unit
- Protein mini-transblot assembly
- Vortex mixer
- Microcentrifuge
- Bunsen burner
- Heat block
- Timer
- Parafilm
- Two waste containers; one biohazard and one nonbiohazard

II. MICROPIPETTING

Micropipettes are tools used to measure extremely small volumes of liquid typically necessary when performing molecular manipulations. We will use four different micropipettes in this course. Each micropipette is accurate to measure a defined range of volume, as shown in the Table 1.1. Your brand of micropipette may vary slightly from the ranges shown, particularly with the lower limit of the P1000 micropipette. Please refer to the manufacturer's instructions.

Setting the micropipettes to the desired volume can be a little tricky at first. Also, it is common for beginners to confuse the P20 and P200 since they typically use the same pipette tips: therefore, remember to check which micropipette you are using before drawing in solution. Many students accidentally measure 20 μL instead of 2 μL or vice versa because of such mix-ups.

When using micropipettes, one should always keep in mind significant figures. The correct number of significant figures is determined by all the numbers visible, plus the volume referred to by the tick mark (found on most brands of micropipettes). Since there are variations between brands of pipettes, most protocols will refer to a volume without indicating the exact number of significant figures. You should always set your pipette to the correct volume, including as many significant figures as possible. For instance, if a protocol instructs you to pipette 5 μL, you should set your pipette to 5.00 μL and carry the correct number of significant figures through any required calculations.

Use Fig. 1.1 and the following instructions to help with setting the micropipettes until you are confident to set them on your own.

Follow these instructions when using the micropipettes:

1. Set the desired volume by holding the pipette in one hand and rotating the dials with the other hand. Do *not* dial past the lower limit (000) or the upper limit (shown on the pipette; either 10, 20, 200, or 1000). Familiarize yourself with these settings.
2. Attach a tip to the end of the micropipette. To ensure an adequate seal, press the tip on with a slight twist.
3. Depress the plunger to the first stop. This part of the stroke displaces a volume of air corresponding to that indicated on the dial.
4. Immerse the tip to a depth of 2–5 mm in the liquid to be withdrawn. Immersing the tip deeper will cause liquid to adhere to the outside of the tip, causing errors in measurement.
5. Allow the plunger to slowly return to its original position. If the plunger snaps back, aerosols will form, contaminating the barrel of the micropipette and your solution.
6. Wait 1 second before removing the tip from the solution, to allow the introduced liquid to enter the pipette tip fully. Removing the tip too quickly from the solution may result in air occupying some of the calibrated volume. Check to

TABLE 1.1

Micropipette	Volume Range (μL)
P10	0.5–10
P20	2–20
P200	20–200
P1000	100 (or 200)–1000

No Permission Required

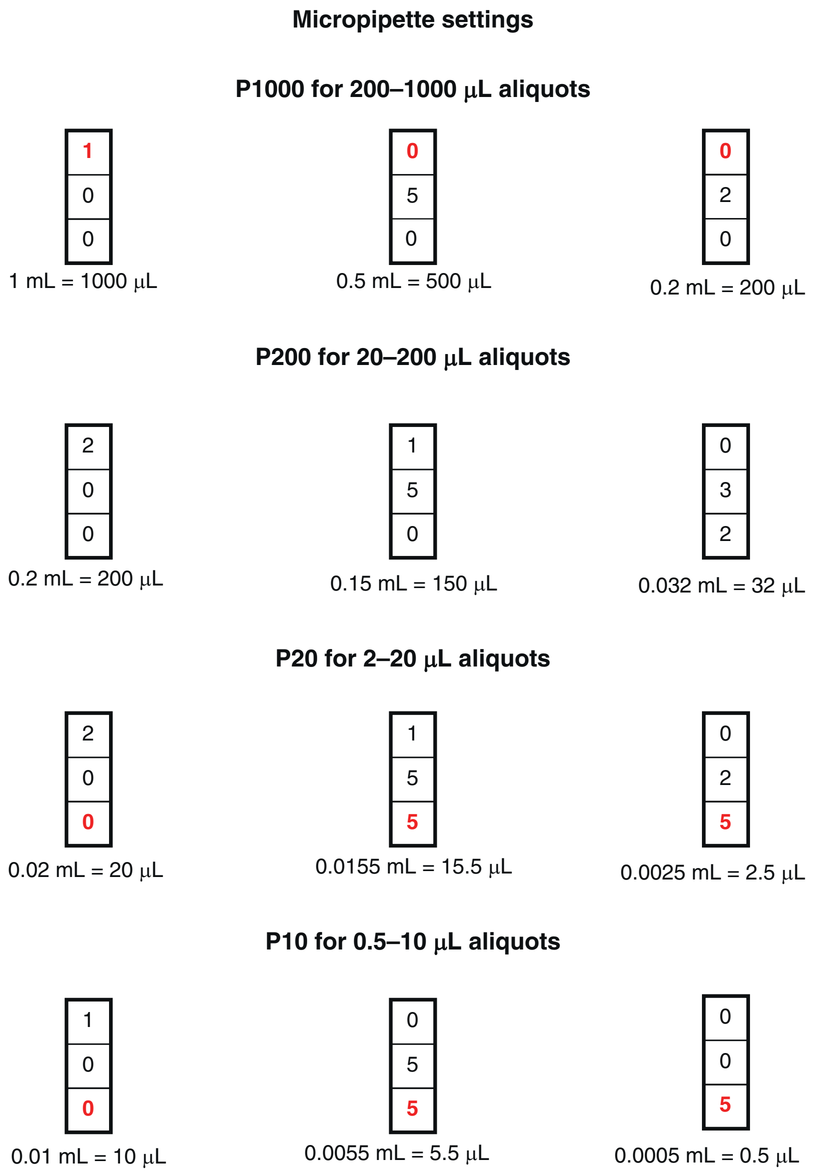

FIGURE 1.1 Micropipette settings cheat sheet. *No Permission Required*

make sure that there are no air bubbles and that the amount of liquid corresponds to the desired amount. Develop an eye for 1 µL volumes, as these are the hardest to pipette.

7. Place the tip against the side wall of the receiving vessel near the liquid interface or the bottom of the vessel. Slowly dispel the contents by depressing the plunger until the first stop. Remaining liquid can be dispelled by depressing the plunger to the second stop. Withdraw the tip from the solution and return the plunger to its original position. Check to make sure that no liquid remains in the tip. If there is a bead of liquid, reintroduce liquid from the receiving vessel to capture the bead and slowly expel the contents.
8. Discard the tip by pressing the ejector button.
9. *Always* use a new pipette tip when pipetting enzymes or the stock solutions may become contaminated. If you accidentally contaminate an enzyme solution, tell an instructor. Always use a new pipette tip for critical volumes, as in a dilution series, because as much as 10% of the volume may stay within the tip after delivery.
10. Working with tiny volumes requires patience and accuracy. The best way to deliver a 1 µL volume is to pick up the receiving tube and make sure that a 1 µL bead is formed on the side of the tube after delivery. In the case of enzymes, schlieren rings should be visible from the glycerol–water interface if the enzyme is dispelled directly into the solution.

III. LABORATORY EXERCISES

Note: While the other laboratory exercises for this course will build on each other, this one will not.

The purpose of this short exercise is to practice using the micropipettes (and test your technique). Each student will perform serial dilutions of the protein bovine serum albumin (BSA) and then compare their results against their lab partner's results using a visualization technique that uses a protein-binding dye.

A. Micropipetting Self-Test

Before proceeding further, each student should do a self-test on their micropipetting skills. Because 1 mL of water weighs approximately 1.00 g (the exact density is 0.99x, with x being temperature-dependent), students can test their micropipetting skills by pipetting into a tared weigh boat on a precision balance. To begin, place a weight boat on a precision balance and tare the balance so it reads 0 g. Next, pipette the appropriate volume of water onto the weigh boat and record the reading. You may tare the scale between volume readings. Continue until you have completed a minimum one set of self-tests.

"Passing" the self-test will ensure that you are selecting the correct micropipette for the given volume, and that your technique is correct. If your self-test does not fall into the right weight ranges, see your instructor for one-on-one feedback about your technique, and to test the calibration of your micropipettes.

Each student should perform at least one set of self-tests; selecting and adjusting their pipettors independently.

Self-Test 1

Volume to Measure (μL)	Weight in g (within 5%)
7.0	0.0070 (0.0067–0.0073)
33.5	0.0335 (0.0318–0.0352)
267.5	0.2675 (0.2541–0.2809)

No Permission Required

Self-Test 2

Volume to Measure (μL)	Weight in g (within 5%)
9.0	0.0090 (0.0086–0.0095)
26.5	0.0265 (0.0252–0.0278)
348.5	0.3485 (0.3311–0.3659)

No Permission Required

Self-Test 3

Volume to Measure (μL)	Weight in g (within 5%)
8.0	0.0080 (0.0076–0.0084)
43.5	0.0435 (0.0413–0.0457)
364.5	0.3645 (0.3463–0.3827)

No Permission Required

Self-Test 4

Volume to Measure (μL)	Weight in g (within 5%)
6.0	0.0060 (0.0057–0.0063)
32.4	0.0324 (0.0308–0.0340)
246.5	0.2465 (0.2342–0.2588)

No Permission Required

B. Preparing Bovine Serum Albumin Dilutions

You will be given a tube with 10 μL of a 10 mg/mL BSA solution. Prepare a dilution series of BSA standards in five tubes (labeled 1–5) according to the following scheme and Fig. 1.2. Attach a new pipette tip to the micropipette each time you make the dilutions. *Each lab partner should do a set of dilutions*.

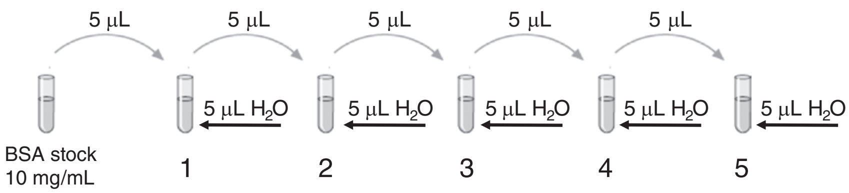

FIGURE 1.2 Serial dilution scheme. *No Permission Required*

Tube	Dilution	Protein Concentration
1	5 μL of stock (10 mg/mL)+5 μL dH_2O *This is a 2-fold dilution*	5.00 mg/mL
2	5 μL from tube 1+5 μL dH_2O *This is a 2-fold dilution*	2.50 mg/mL
3	5 μL from tube 2+5 μL dH_2O *This is a 2-fold dilution*	1.25 mg/mL
4	5 μL from tube 3+5 μL dH_2O *This is a 2-fold dilution*	0.63 mg/mL
5	5 μL from tube 4+5 μL dH_2O *This is a 2-fold dilution*	0.31 mg/mL

No Permission Required

C. Performing a Nitrocellulose Spot Test

Ponceau Red is a stain that quantitatively binds a protein. You will use a micropipette to deliver small amounts of BSA serial dilutions to the nitrocellulose membrane and then stain the nitrocellulose. This will enable you to visualize the relative protein amounts in each sample and provide visual feedback on your pipetting/dilution technique.

1. Obtain a piece of nitrocellulose (always wear gloves when handling nitrocellulose). Place the nitrocellulose on a piece of 3MM paper (Whatman, Clifton, NJ). You will share the piece of nitrocellulose with your partner.
2. Spot 2 μL volumes of deionized H_2O (dH_2O; control) and each of the BSA dilutions. One partner should spot the top row with their samples, and the other partner should spot a row below.
   Spotting of the 2 μL is best done by holding the pipette tip just above the paper. Expel liquid such that a drop forms on the end of the tip. Touch the drop to the paper and the liquid will be drawn into the paper by capillary action.
   CAUTION: Make certain that you leave enough room between each addition so that the spots do not touch each other.
3. After the spots have dried completely, stain the nitrocellulose membrane by placing it in a tray (a square Petri dish works well for this purpose) and adding enough Ponceau Red staining solution to cover the membrane. Allow the stain to incubate for 1 to 2 minutes, with gentle rocking/shaking.
4. Pour off the stain (into a bottle labeled "Used Ponceau Red") and rinse the membrane three times with dH_2O. Add enough water to cover the membrane and rock/shake gently on a platform rocker or an orbital shaker. Change to fresh dH_2O after 5 minutes and rock/shake gently until the background is white. *Note: Ponceau Red can be reused multiple times, but eventually will stop working as it becomes diluted.*
5. Place the nitrocellulose on 3MM paper to dry. Compare the intensities of each spot. Do the intensities of your spots match your lab partner's? Does each spot appear to be half as intense as the last? If not, you need to practice your micropipetting technique.

DISCUSSION QUESTIONS

1. What are some real-life applications of biotechnology? What are some important recombinant proteins and/or recombinant organisms that are used today?
2. What are your goals in taking this class? What are you hoping to learn, and how do you hope it will expand your career or future research?